

Strategy Evaluation Report

CS7646

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1.1 INTRODUCTION

In this report, we will see different ways stock indicators can be used to implement trading strategies. We use a Manual Strategy and a Strategy Learner that buys and sells based off three different indicators: Simple Moving Average, Momentum, and Bollinger Bands Percentage. For the Strategy Learner I decided to use a classification-based learner which uses 20 bags of Random Tree Learners with a leaf size of 5.

1.2 INDICATOR OVERVIEW

Indicator 1: Simple Moving Average

The first indicator I used was the Simple Moving Average (SMA) indicator. This is a moving average of the prices for a given period. The SMA is calculated using the following equation below.

$$SMA_i = \frac{\sum_{k=i-n}^i x_k}{n}$$

To implement this, I calculated this indicator by finding the rolling mean of the adjusted closing prices. The parameter that I had to adjust for this project was the window size and found that a window size of 10 worked very well.

Indicator 2: Momentum

The second indicator I used was the Momentum indicator. The Momentum indicator is mainly used to figure out the strength/weakness of a stock. It

compares the current stock price with the stock price from a different period. The momentum is calculated using the following equation below.

$$Momentum = \frac{PriceClose(i)}{PriceClose(i - n)} \cdot 100$$

To implement this, I calculated the momentum by finding the current adjusted closing price and dividing it by the adjusted closing price from 10 days before the current day. The parameter I had to tweak to optimize this was the window size, which gave a good lookback period for the Manual Strategy and Strategy Learner.

Indicator 3: Bollinger Bands Percentage

The third indicator I used was the Bollinger Bands Percentage (BB%) indicator. The BB% indicator uses the Simple Moving Average (SMA) and it is calculated by finding the upper bound and lower bound by calculating 2 standard deviation above and below the SMA.

To implement this, I had to find the Bollinger Bands Percentage, by subtracting the prices by the lower band and dividing that by the upper band minus lower band. I initially had to Bollinger Bands as my indicator, but had to change it to work with the Manual Strategy and the Strategy Learner by making it return just the Bollinger Bands Percentage. I had to tweak the window size parameter and once again found that 10 was a suitable window size.

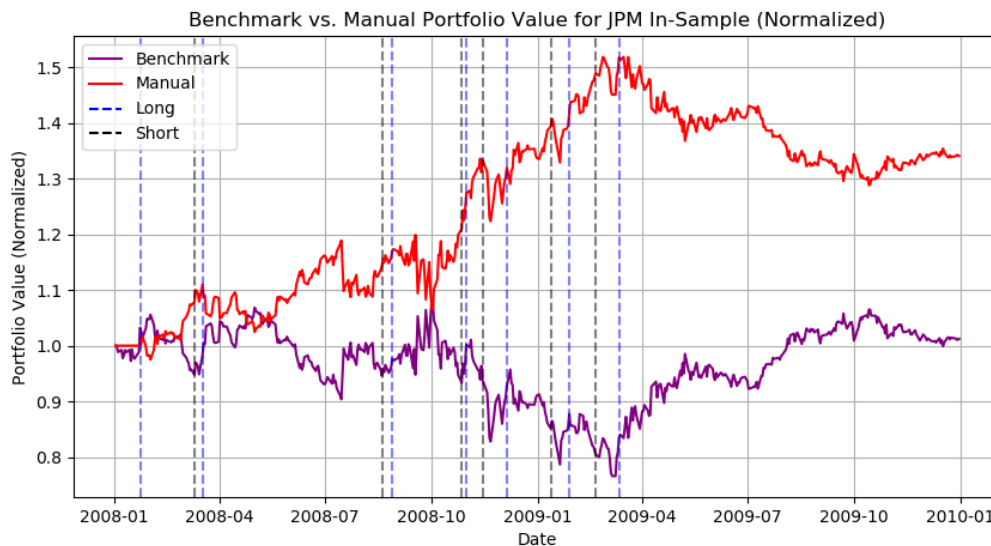
1.3 MANUAL STRATEGY

For the Manual Strategy, I had to figure out how to combine the three indicators so that the strategy can work properly. To do this, I found when each indicator has a buy signal and a sell signal. I learned that the Simple Moving Average indicator has a buy signal when the SMA is greater than the current days price,

and inversely a sell signal when the SMA is less than the current days price. For Momentum I found that when it is less than either 0 or -.15 then that indicates a buy signal, and when it is greater than 0 or -.15 then that indicates a sell signal. For this strategy I set the threshold at -.15 for momentum. For Bollinger Bands Percentage, I found that when BB% is below .2 that there is a buy signal, and when BB% is above .8 that there is a sell signal.

To decide whether to enter or exit my positions I had to think of a way to implement all three of these indicators together to determine whether to go long or short. My final strategy was just to simply only buy when all 3 indicators indicated a buy signal and only sell when all 3 indicators indicated a sell signal. I took the SMA, Momentum, and BB% for everyday and compared them with their respective thresholds and from that, is how I decided whether to enter or exit the current position.

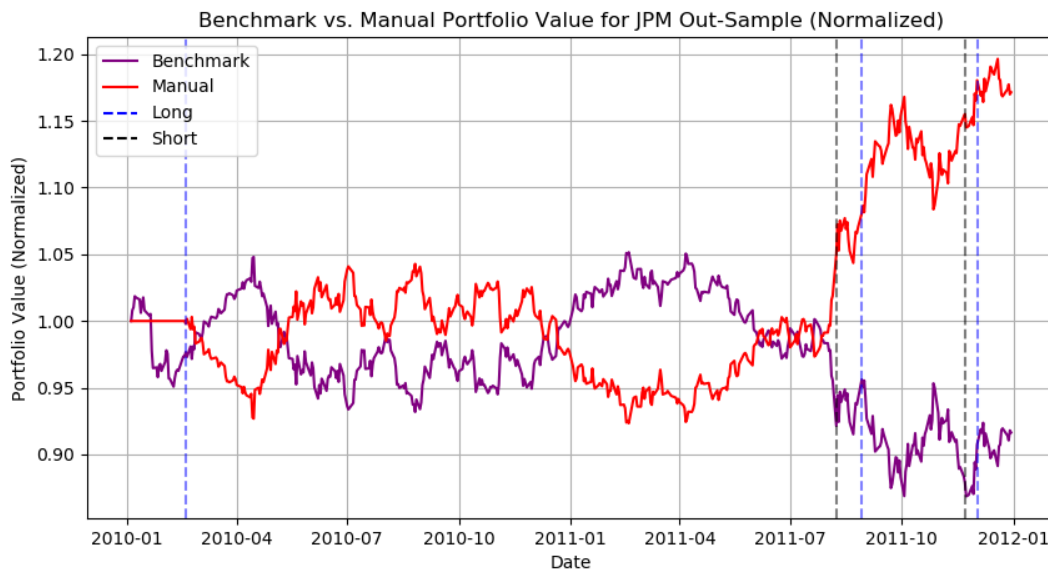
I believe this is an effective strategy because you must meet all 3 indicators criteria before the strategy enters or exits a position. This helps ensure that opposite does not happen such as what could happen a lot if you just used one or two indicators. This also helps there not be too many trades that happen, and it only trades when there is a very high signal.



Above we can see the Manual Strategy compared to the Benchmark which is just if you bought 1000 shares of JPM on January 1st, 2008, and held onto that until December 31st, 2009. We can see where the Manual Strategy went long and short by the blue and black lines respectively.

	Benchmark	Manual Strategy
Cumulative Returns	0.012325	0.340811
STD of Daily Returns	0.017041	0.012805
Mean of Daily Returns	0.000169	0.000664

From this chart above, we can see just how much the Manual Strategy compared to the benchmark. We see that there is a drastic difference in cumulative return, with the Manual Strategy having 34.08% increase in cumulative return.



Above we can see the Manual Strategy compared to the Benchmark from January 1st, 2010, and held onto that until December 31st, 2011. This time, we can see in the out-sample that there was not as many long and short position changes than in the in-sample.

	Benchmark	Manual Strategy
Cumulative Returns	-0.083579	0.171382
STD of Daily Returns	0.0085	0.007487
Mean of Daily Returns	-0.000137	0.000342

Above, we can see the following stats from the out-sample results. The benchmark actually had a negative return, whereas the Manual Strategy had a cumulative return of a 17.14% increase.

From the out-sample period, I notice that there are entrances to long and short positions. That could be due to the differing nature of the market when compared to the in-sample period. Because of this difference, we can see times where the Manual Strategy is actually performing worse than the Benchmark, and that is because with my strategy there was only 1 long entry point near the beginning of the time period at around February 2010. We find that the cumulative return in the in-sample period was greater than the cumulative return in the out-sample.

1.4 STRATEGY LEARNER

For the Strategy Learner, I used a Bag Learner with a bag size of 20 which consist of Random Tree Learners that have a leaf-size of 5. This is a classification learner, so to classify, I converted Buy, Sell, and Hold to 1, -1, and 0 respectively in addEvidence(). From these values, the learner will know whether to Buy, Sell, or Hold. The train_x data is all 3 of the indicators I used stored together in one dataframe. From the train_x data, we are able to get the train_x data which is either 1, -1, or 0. In testPolicy() I get the test_x which is all the indicator information together, and then query for the test_y. From this I create a trades dataframe that can then be used by the compute_portvals function which is in marketsimcode.

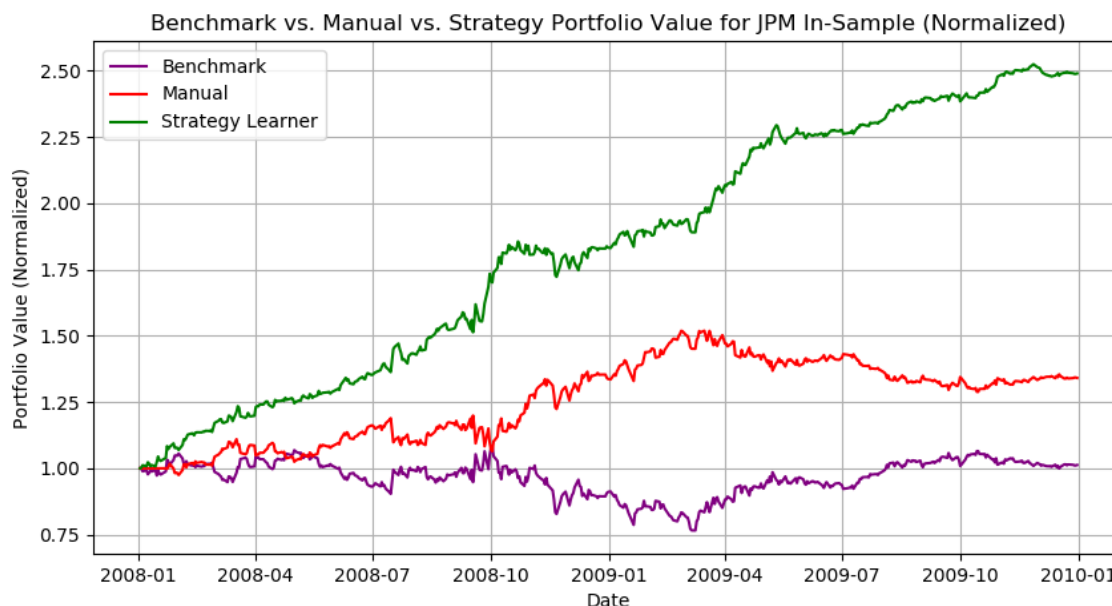
There was no need to discretize or adjust my data since I used a Random Tree Learner with the Bag Learner. The only adjustments I had to make was concatenating my indicator data together so that it can be used in the Strategy Learner. I also just had to make the change of using mode inside my learners

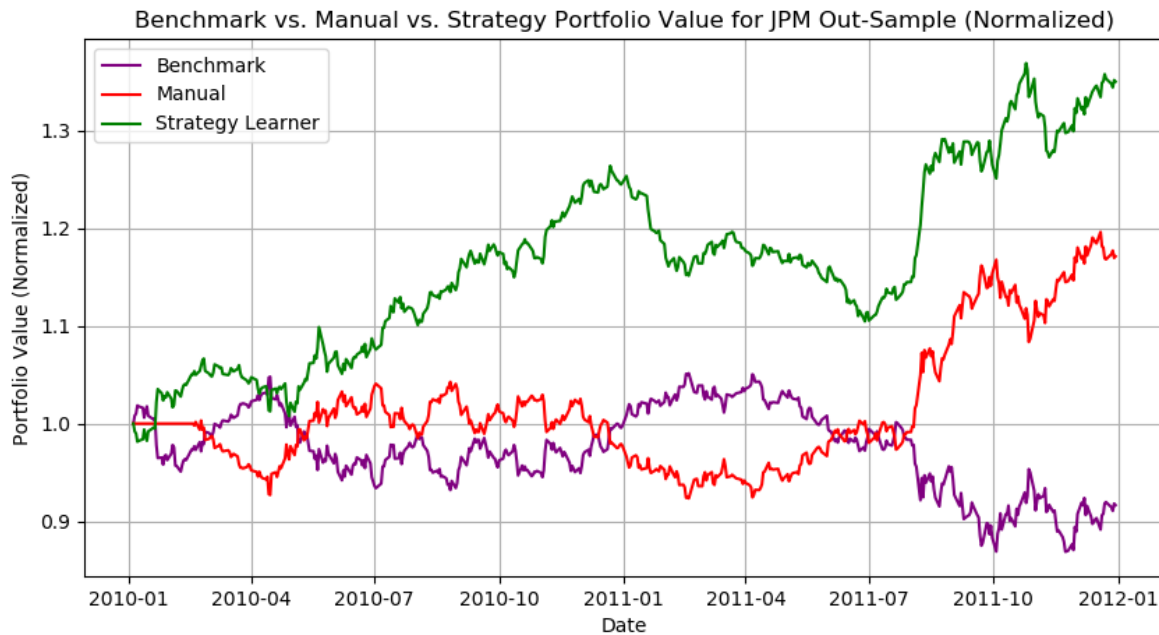
because we are using classification, and taking the mean would not help classify based off of 1, -1, 0.

1.5 EXPERIMENT 1 (MANUAL STRATEGY / STRATEGY LEARNER)

For this experiment, I used the dates of January 1st, 2008 through December 31st, 2009 for the in-sample period, and the dates of January 1st, 2010 through December 31st, 2011 for the out-sample period. Both use the transaction costs of Commission: \$9.95 and Impact: 0.005. Also, all this is performed on the symbol "JPM".

For this experiment we will compare the benchmark portfolio value against the Manual Strategy and the Strategy Learner for the in-sample and out-sample period. For the in-sample period, I hypothesize that the Strategy Learner will vastly outperform the benchmark and Manual Strategy since it is being trained on the data. For the out-sample, I hypothesize that the Manual Strategy and Strategy Learner will be near similar in cumulative return, but the Strategy Learner might be a bit better.





In the above charts, you can see the performance for the Manual Learner, Strategy Learner, and the Benchmark for the in-sample and out-sample periods. In the in-sample chart, we can see that exactly what I hypothesized ended up being true. The Strategy Learner outperformed both the Manual Strategy and the Benchmark heavily. I would expect this relative result for the in-sample data every time because the Strategy Learner is trained on the in-sample period data, thus you would expect the test results to be great.

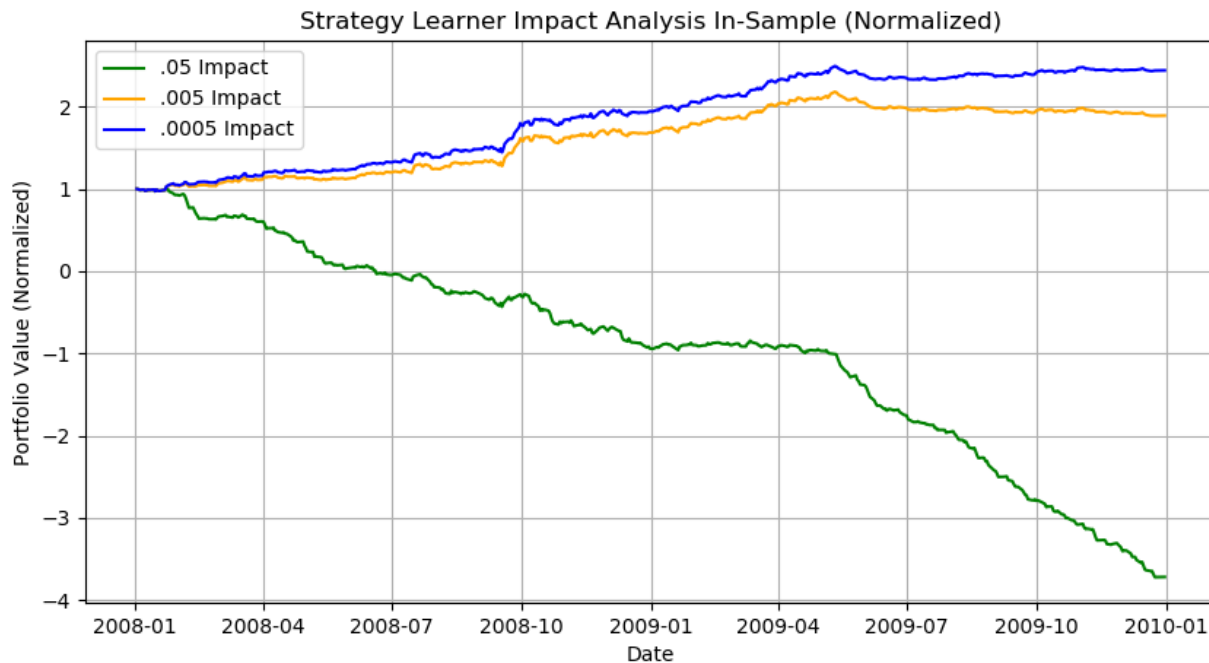
The out-sample chart sort of met my hypothesis. I knew that it would not perform as good as the in-sample since it is not trained on that period. It still performed close to the Manual Strategy, which is something that was expected. From this we can see that the Manual Strategy does well no matter if it is in the in-sample or out-sample, and we also see that the Strategy Learner does very well on in-sample data compared to out-sample.

1.6 EXPERIMENT 2 (STRATEGY LEARNER)

In this experiment we will be analyzing how different impact values affect in-sample trading behavior and results with the Strategy Learner. Impact is the

change in the price of an asset caused by trading that asset. For this experiment we will setup 3 Strategy Learners, all using 20 bags of Random Tree Learners with a leaf size of 5. We will be using the in-sample period of January 1st, 2008 to December 31st, 2009. We will be setting the commission to 0 for each Strategy Learner. Each learner will have a different impact, one for 0.05, 0.005, and 0.0005.

My hypothesis for this experiment is that when impact is high, we will see lower cumulative returns, and when the impact is low, we will see higher cumulative returns. This is because when impact is low, the cost towards trading the stock does not affect it as much as when the impact is higher.



In the above chart, you can see the portfolio value for the Strategy Learners with differing impacts of 0.05, 0.005, and 0.0005. These results met my hypotheses expectations because from the chart we can see the portfolio value of the Strategy Learner with the highest impact (0.05) performed the worse and we can see that the portfolio value of the Strategy Learner with the lowest impact (0.0005) performed the best of the three.

	Impact: 0.05	Impact: 0.005	Impact: 0.0005
Cumulative Returns	-4.719632	0.8918	1.442256
STD of Daily Returns	0.216348	0.010359	0.009568
Mean of Daily Returns	-0.001798	0.001319	0.001819

The following stats above were calculated for the 3 different learners with varying impacts. We can see that my hypothesis was correct from this since the learner with 0.05 impact had a -47.19% cumulative return compared to the lower impact learners, 0.005 and 0.0005, having 89.18% and 144.23% respectively. We can also look into the average daily returns on these learners and see that the 0.05 impact learner had a -.1798% average daily return, the 0.0005 impact learner had a .1819% average daily return and the 0.005 impact learner was in between both of the others with a average daily return of .1319%.